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Idea Man: A Memoir by the Co-founder of Microsoft

by Paul Allen

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93 Highlights | 2 Notes

Highlight (Yellow) | Location 64

Over grinders or a pepperoni pie at the Harvard House of Pizza, we fantasized about our entrepreneurial future. One time I asked Bill, “If everything went right, how big do you think our company could be?” He said, “I think we could get it up to thirty-five programmers.” That sounded really ambitious to me.

Highlight (Yellow) | Location 67

In 1965, in a journal called Electronics, a young research physicist named Gordon Moore made that prediction specific. He asserted that the maximum number of transistors in an integrated circuit would double each year without raising the chip’s cost. After cofounding Intel in 1968, Moore amended the rate of doubling to once every two years—still dramatic. Similar trends soon emerged in computer processing speed and disk storage capacity. It was a simple but profound observation that holds true to this day. Because of continual advances in chip technology, computers will keep getting markedly faster and cheaper. The momentum of Moore’s law became more evident in 1969, a few months after I’d met Bill.

Highlight (Yellow) | Location 92

My really big ideas have all begun with a stage-setting development—in this case, the evolution of Intel’s early microprocessor chips. Then I ask a few basic questions: Where is the leading edge of discovery headed? What should exist but doesn’t yet? How can I create something to help meet the need, and who might be enlisted to join the crusade? Whenever I’ve had a moment of insight, it has come from combining two or more elements to galvanize a new technology and bring breakthrough applications to a potentially vast audience. A few months after the 8008 was announced, one of those brain waves came to me. What if a microprocessor could run a high-level language, the essential tool for programming a general-purpose computer?

Highlight (Yellow) | Location 105

Bill and I had already found a groove together. I was the idea man, the one who’d conceive of things out of whole cloth. Bill listened and challenged me, and then homed in on my best ideas to help make them a reality. Our collaboration had a natural tension, but mostly it worked productively and well.

Highlight (Yellow) | Location 216

I was given an oversize picture book with everything a four-year-old could want to know about steam shovels, tractors, backhoes, and fire engines. I read that book every day. Seeing my interest, my mother had a friend give me a tutorial on steam engines. It wasn't very technical, but I got my first inkling about the gears and belts and all the other hidden things that make a machine come alive. That book opened a new world to me. Soon I was pleading for one on gasoline engines. Later I progressed to steam turbines and eventually to atomic power plants and rocket engines. I'd pore over each volume, not getting all the details but grasping enough to satisfy me. On some elemental level, the magical became logical. I began to understand how these things worked.

Highlight (Yellow) | Location 235

Each morning she would send me into the world with a paraphrase of the Spartan mothers' farewell to their sons marching off to war: "Go forth bearing your shield!" I walked out the door a little straighter when I heard that.

Highlight (Yellow) | Location 277

"When you grow up and have a job, do something you love. Whatever you do, you should love it." He'd repeat this to me over the years with conviction. Later I'd figure out what he meant: Do as I say, not as I've done. Much later, my mother told me that my father had wrestled with his career choice. He suspected he might be happier coaching football than managing libraries, but he finally chose the safe and practical route, a nine-to-five life under fluorescent lights. Lots of men from his generation did the same. But he wanted me to choose better.

Highlight (Yellow) | Location 281

THE OFFICIAL GOAL of the 1962 Seattle World's Fair was to inspire young people to pursue careers in science. The unofficial goal was to show that the United States had caught up to the Soviet Union in technology and the space race. But for me, a nine-year-old who'd just discovered science fiction, the Century 21 Exposition (as it was officially titled) revolved around my favorite thing: the future. It was like waking up to find my most outlandish ideas made real, just four miles from my house.

Highlight (Yellow) | Location 297

my first Belgian waffle, which seemed like the most exotic and delicious thing I'd ever eaten. You can catch a close-up of that waffle in an Elvis Presley movie called It Happened at the World's Fair: an oversize, crispy square slathered with whipped cream and topped with sliced strawberries and powdered sugar.

Highlight (Yellow) | Location 314

Midway through the meal, we heard a strange noise, somewhere between a wheeze and a choking rasp. What was that? Back to our food and talk, we heard it again, louder this time, clearly coming from downstairs. I trailed behind Dad, who pushed open the basement door. There was Jett, quivering at the top of the stairs. At the bottom it looked like a foggy morning on the Okefenokee, with two feet of yellow-green chlorine gas blanketing the floor. Jett had made the smart move to get as far as possible from the toxic fumes. As my father raised the

basement windows to clear out the gas, he said, “You’ve got to be more careful with your experiments, Paul.” But I also heard what he didn’t say: He never told me to stop. In the Allen household, children were treated like grown-ups. Our parents encouraged us at whatever we tried, and exposed us to Bach and jazz and flamenco, but it was more than that. They respected us as individuals who needed to find our own place in the world.

Highlight (Yellow) | Location 401

My second big moment came when I was chosen to deliver the graduation address for Lakeside’s lower school. It was my first speech, and I slaved over it. As I rose before classmates, faculty, parents, and honored guests, I felt a strange sensation in my legs. My knees were knocking, just like a cartoon. It was 1967, and artificial intelligence was the hot theme in science fiction. I’d read Isaac Asimov’s *I, Robot*, with its First Law of Robotics (“A robot may not injure a human being or, through inaction, allow a human being to come to harm”), and *Colossus*, a 1966 British novel about a malevolent megacomputer that wound up ruling the world. Newspapers of the day were filled with headlines like “Computers Are Taking Over,” or “Automated Government Is Here.” I began by hailing “the age of the computer” and a future that “holds for us the bright prospect of even more remarkable things to come.” After acknowledging the specter of computers someday replacing human workers on assembly lines, I paid my respects to the machines’ “amazing capabilities”

Highlight (Yellow) | Location 500

One day early that fall, I saw a gangly, freckle-faced eighth-grader edging his way into the crowd around the Teletype, all arms and legs and nervous energy. He had a scruffy-preppy look: pullover sweater, tan slacks, enormous saddle shoes. His blond hair went all over the place. You could tell three things about Bill Gates pretty quickly. He was really smart. He was really competitive; he wanted to show you how smart he was. And he was really, really persistent. After that first time, he kept coming back. Many times he and I would be the only ones there.

Highlight (Yellow) | Location 505

remember the first time I went to Bill’s big house a block or so above Lake Washington, feeling a little awed. His parents subscribed to *Fortune* and Bill read it religiously. One day he showed me the magazine’s special annual issue and asked me, “What do you think it’s like to run a Fortune 500 company?” I said I had no idea. And Bill said, “Maybe we’ll have our own company someday.” He was thirteen years old and already a budding entrepreneur.

Highlight (Yellow) | Location 521

peered through the plate glass, into a room that never went dark, at the mysterious puppy in the window: a black mainframe with cabinet after cabinet and panels of blinking lights. The CPU alone was about five feet wide. It was the first time that I’d seen an actual computer in the flesh, and it seemed not quite real that such a thing could exist just forty blocks from where I lived. All I wanted to do at that moment was log on, connect, and have at it. Today’s average laptop is thirty thousand times faster than the machine I was lusting after, with ten thousand times more memory. But in its day, the PDP-10 was the most advanced species of an evolutionary alternative to the batch-processing establishment.

Highlight (Yellow) | Location 559

way to learn was going hands-on with what was the top machine of the time, learning about how it worked, what it took to ‘make it or break it.’” Another approach was to stress-test a piece of software until it failed,

Highlight (Yellow) | Location 564

We adopted their jargon; a kludge, for example, was a baling-wire-and-gum sort of coding fix.

Highlight (Yellow) | Location 574

the Giant Bryant. The name stuck. The drive was built to heroic dimensions. A massive electric motor at the center ran a thick shaft that supported a dozen oxide-coated steel disks, each more than three feet in diameter. They spun in unison while a set of hydraulic arms with magnetic heads, floating on thin cushions of air, moved across their surfaces to read the data. The drive could store around 100 million characters, an order of magnitude beyond anything else available. (Today’s laptop drives typically store six hundred times as much data in 0.002 percent of the volume.)

Highlight (Yellow) | Location 607

For me, the Holy Grail of software was the operating system, the computer’s nervous system. It does the logistical work that allows the central processing unit to compute: shifting from program to program; allocating storage to files; moving data to and from modems and disk drives and printers. You don’t think about it unless something goes wrong and it crashes.

Highlight (Yellow) | Location 639

Lakeside schoolmate once said that I could read assembly code the way other people read novels, but I don’t think it came more easily to me. It was simply the place I chose to focus, where the rubber met the road. I was finally beginning to understand how a computer worked at its most fundamental level. I’d made it into the guts of the machine.

Highlight (Yellow) | Location 691

Finally they got to us and said, “OK, boys, we’ve got to take the chairs.” We knelt on the floor by our terminals to finish saving our programs. Minutes later, I saw Bill staring out the window with his mouth open. One of the wheeled chairs had gotten loose and was rolling down Roosevelt Way with a moving man giving chase. We burst out laughing, but it was gallows humor, the end of a vital phase of our lives. My prospects for a summer job were gone, and my father was threatening to revoke my driving privileges if I didn’t make up some missing chem labs. More seriously, the bankruptcy was a lasting reminder of how a business can plummet in a flash. Bill, in particular, never forgot that lesson.

Highlight (Yellow) | Location 733

I was driven more by curiosity than by the compulsion to get good grades.

Highlight (Yellow) | Location 774

but college life wasn't quite what I'd hoped. My intro courses were less than challenging. I missed my family and my girlfriend and got distracted by the social hubbub. Some people thrive when they first go out on their own, but I was not one of them.

Highlight (Yellow) | Location 933

For the most part, the two of us got along well that spring and summer. But at times Bill could get edgy, especially during our chess games. I was a more methodical player, my openings more structured; Bill was an aggressive improviser. When I beat him one day, he got so angry that he swept the pieces to the floor. "That was the stupidest move I ever made!" Bill shouted.

Highlight (Yellow) | Location 943

My first task was to define a set of thirty or so "macros," the symbolic instructions that would generate bytes for the Intel 8008. Within a few days, I'd effectively performed a brain transplant. The PDP-10's assembler didn't know it, but it was now an 8008 assembler. My next step was to build the simulator itself, a program to put this metamorphosis into action. Written in PDP-10 assembly language, the simulator would mimic the microchip's instructions. The coding went smoothly; it was as though everything I'd learned at C-Cubed and ISI had led me to this point. Fortunately, I was able to finish my week's work for Bob Barnett in about twenty hours, then focus on Traf-O-Data. After going hard for a week, I was done. My third and final step was to modify the PDP-10's debugger to give Bill the ability to stop the program in midexecution and track the cause of any problem. The debugger was an ugly, hairy piece of code, full of trapdoors and cul-de-sacs, but three weeks after I'd begun, we had an unrivaled development suite for the 8008 chip. (My techniques proved so effective that Microsoft used them well into the 1980s, until microprocessors became fast and capable enough to host their own development tools.)

Highlight (Yellow) | Location 958

Another time, as Bill and I dined at a local pizza place, I had a thought: "What if you could get the news by reading from a computer terminal instead of buying a newspaper? You could even program it to find articles on whatever you wanted. Wouldn't that be great?" And Bill said, "Come on, Paul! It costs seventy-five dollars a month to rent a Teletype and you can get a paper delivered for fifteen cents. How do you compete with that?" He had me there. But I couldn't stop thinking about a time when everyone would be digitally connected, with instant access to information and services. It would be a while before Bill and I defined our goal, in so many words, as "a computer on every desk and in every home." But the seeds of that motto—and my notion of a global network to join all those computers together—would be planted that summer among the strip malls and fast-food joints in Vancouver, Washington.

Highlight (Yellow) | Location 971

moments when you realize, I just can't see it. I felt a little sad, but I accepted my limitations. I was OK with being a generalist. For Bill it was different. When

Highlight (Yellow) | Location 1001

Above all, we learned that it was hard to compete with “free.” (Bill took that lesson to heart. Years later he’d become obsessed with Linux, the open-source operating system.) But there were positives, too. Traf-O-Data bolstered my conviction that microprocessors would soon run the same programs as larger computers at a far lower cost. Looking ahead, my development tools for the 8008 would give us an invaluable foundation when that next-generation chip came along.

Highlight (Yellow) | Location 1007

In my experience, each failure contains the seeds of your next success—if you are willing to learn from it. Bill and I had to concede that our future wasn’t in hardware or traffic tapes. We’d have to find something else.

Highlight (Yellow) | Location 1025

as I started at Honeywell, the company that made the thermostats in my childhood home.

Highlight (Yellow) | Location 1040

“Did you see that?” she said, after he’d left. “He ate his chicken with a spoon. I have never in my life seen anyone eat chicken with a spoon.” When Bill was thinking hard about something, he paid no heed to social convention. Once he offered Rita fashion advice—basically, to buy all your clothes in the same style and colors and save time by not having to match them. For Bill, that meant any sweater that went with tan slacks.

Highlight (Yellow) | Location 1049

spent more time with Bill at Currier House before his nightly poker games with the local cardsharps. He was getting some costly lessons in bluffing; he’d win three hundred dollars one night and lose six hundred the next. As Bill dropped thousands that fall, he kept telling me, “I’m getting better.” I knew what he was thinking: I’m smarter than those guys.

Highlight (Yellow) | Location 1065

SOME HAVE SUGGESTED that our Altair BASIC was remarkable because we created it without ever seeing an Altair or even a sample Intel 8080, the microprocessor it would run on. What we did was indeed unprecedented, but what is less well understood is that we had no choice. The Altair was little more than a bare-bones box with a CPU-on-a-chip inside. It had no hard drive, no floppy disk, no place to edit or store programs. And even had the machine been up to it, debugging on the memory-challenged 8080 would have been slow and difficult at best. Any other programmers vying to bring an 8080 BASIC to Albuquerque would be facing an uphill climb. For starters, they’d have to realize that they needed a simulator and then to create one from scratch on a mainframe or minicomputer. Bill and I had a big edge in speed and productivity with our Traf-O-Data development tools. But could we actually write a BASIC interpreter?

Highlight (Yellow) | Location 1112

At one point, after Monte laid his floating-point math routine on the floor, Bill sprawled next to the long tail of fan-folded paper to find a few last trims. We knew that each byte saved would leave that much more room for users to add to their applications. (Today we live in a different world, where sixteen gigabytes of memory—four million times our BASIC budget for the Altair—are packed into the base iPhone. Handcrafted code is by and large a lost art.

Highlight (Yellow) | Location 1121

As we pushed on, our confidence grew. One day we called MITS to ask about the handshake subroutines on the Teletype for coding the Altair's input and output. We knew we were on the right track when Bill Yates, Ed Roberts's partner and lead engineer, told us that no one else had even posed the question. At that point we figured the job was ours to lose.

Highlight (Yellow) | Location 1125

we didn't know that Harvard split the computer's maintenance costs with the U.S. Defense Department, based on usage. I'd relied on Bill's password account for my work on the simulator, which ate a lot of processor time. When the January bills came due, Harvard's share was up conspicuously, with one student the prime culprit: William Henry Gates III.

Highlight (Yellow) | Location 1128

By late February, eight weeks after our first contact with MITS, the interpreter was done. Shoehorned into about 3,200 bytes, roughly two thousand lines of code, it was one tight little BASIC—stripped down, for sure, but robust for its size. No one could have beaten the functionality and speed crammed into that tiny footprint of memory: “The best piece of work we ever did,” as Bill told me recently. And it was a true collaboration. I'd estimate that 45 percent of the code was Bill's, 30 percent Monte's, and 25 percent mine, excluding my development tools.*

Highlight (Yellow) | Location 1196

The acid test would be a standard command that we'd used as a midterm exam for our software back in Cambridge. It relied on Bill's core coding and Monte's floating-point math and even my “crunch” code, which condensed certain words (like “PRINT”) into a single character. If it worked, the lion's share of our BASIC was good to go. If it didn't, we'd failed. I typed in the command: PRINT 2+2 The machine's response was instantaneous: 4 That was a magical moment. Ed exclaimed, “Oh my god, it printed ‘4!’” He'd gone into debt and bet everything on a full-functioning microcomputer, and now it looked as though his vision would come true. He couldn't get over the fact that Bill and I had solved the puzzle without any of the hardware—that was astonishing to him.

Highlight (Yellow) | Location 1283

AFTER HARVARD LET out for the summer, Bill and Monte joined me in Albuquerque. We rented a furnished two-bedroom apartment on the ground floor of the Portals, a five-minute drive from MITS. It was a standard medium-rise apartment building, with shag carpeting and a courtyard swimming pool that we never used. Later we added Chris Larson, a younger Lakeside student who'd originally been conscripted for Traf-O-Data. Bill and I each took a bedroom, while Monte and Chris made do with the couch or floor. In need of transportation, I bought my first new car, a metallic blue hatchback Chevy Monza. Bill came with me to pick it up, and we had a hilarious time getting it home. I'd never used a stick shift before, and I'd pop the clutch and stall every twenty feet. Bill tried, with similar results. The Monza was a high-powered little number with a V-8 engine and an undersize clutch that I'd burn out once a year.

Highlight (Yellow) | Location 1395

Fed up, Bill penned "An Open Letter to Hobbyists" for the February 1976 issue of Computer Notes. Identifying himself as "General Partner, Micro-Soft" (likely the first published use of the name), he explained how we'd developed our BASIC and how our royalties thus far amounted to less than two dollars per hour of our time. Why is this? As the majority of hobbyists must be aware, most of you steal your software. Hardware must be paid for, but software is something to share. Who cares if the people who worked on it get paid? Is this fair?... One thing you do [by stealing software] is prevent good software from being written. Who can afford to do professional work for nothing?... The fact is, no one besides us has invested a lot of money in hobby software.... Most directly, the thing you do is theft. Although Bill had gone no further than David or Ed, he provoked a much angrier reaction. Maybe it was his stinging sarcasm, or the fact that David got Bill's letter published in half a dozen hobbyist periodicals. The Southern California Computer Society threatened a class action suit for defamation. Jim Warren, editor of Dr. Dobb's Journal of Computer Calisthenics, wrote: There is a viable alternative to the problems raised by Bill Gates in his irate letter to computer hobbyists concerning "ripping off" software. When software is free, or so inexpensive that it's easier to pay for it than to duplicate it, then it won't be "stolen." Though we received no checks from repentant pirates, the debate became part of a gradual climate change. While theft remained common, both end users and commercial firms began to accept that software, like hardware, had intrinsic value.

Highlight (Yellow) | Location 1429

The star panelist was Ted Nelson, the wild-eyed author of the underground sensation Computer Lib. (Nelson also co-invented hypertext, by which words in one article can link to a related page, a pervasive aspect of today's Web.) He summed up his wild philosophy in four maxims: "Most people are fools, most authority is malignant, God does not exist, and everything is wrong."

Highlight (Yellow) | Location 1455

Later that year, we made our first sales to third-party manufacturers: Data Technology Corporation, National Cash Register, Citibank, General Electric. All were for flat fees, which locked us in as the exclusive provider of BASIC for their machines. Our strategy was to price our products so low that it wouldn't pay for hardware companies to develop their own BASIC, especially since it would delay their entry into the market to do so. But

after a \$50,000 sale to GE, Bill had seller's remorse. As he wrote to me: I think we got bargained down to a very low level in the GE deal and unless we get the [royalty] protection we need in that, I don't think it is a good deal for MITS or us.... The main idea is to get a good deal that doesn't require much work for as much money as possible without being too inconsistent with other sales of BASIC....

Highlight (Yellow) | Location 1462

A short time later, we licensed BASIC to NCR for \$175,000. Even with half the proceeds going to Ed Roberts, that single fee would pay five or six programmers for a year. Though we still relied on MITS's marketing team to help with OEM sales, our business was growing fast. Over time, as Microsoft became the language development company for the personal computer industry, its partners divided the labor to play to their strengths. Bill focused on legal and contract issues, drummed up new business, and navigated our license sales. Whenever I pointed him to a new microcomputer, he'd be all over it to try to sell our software.

Highlight (Yellow) | Location 1469

Where Bill eyed tomorrow's markets, I looked to a more distant horizon. What would our customers want six months or a year from now, and what did we need to do to get it to them before anyone else? Our primary task was to adapt BASIC, our bread-and-butter product, for a new group of microprocessors now competing with the Intel 8080. This work had kicked off the previous fall, when MITS announced an Altair for Motorola's new 8-bit chip. It gathered steam with the release of the 6502 chip from MOS Technology at an unheard-of \$25. For each new microprocessor, I created a new set of development tools on the PDP-10, while Ric helped with the BASIC interpreter rewrites. Though the work was a grind, I was glad to extend our company's reach. Bill and I aimed to provide all language software for every microcomputer on the market. Those BASIC adaptations would be the bedrock of our revenue for years to come.

Highlight (Yellow) | Location 1508

Bill knew that I would balk at a two-to-one split, and that 64 percent was as far as he could go. He might have argued that the numbers reflected our contributions, but they also exposed the differences between the son of a librarian and the son of a lawyer. I'd been taught that a deal was a deal and your word was your bond. Bill was more flexible. In my experience, he believed that agreements were open to renegotiation until they were signed and sealed. There's a degree of elasticity in any business dealing, a range for what might seem fair, and Bill pushed within that range as hard and as far as he could.

Highlight (Yellow) | Location 1632

But no one was more idiosyncratic than Gordon Letwin, a brilliant nerd's nerd who would lock himself in his office and generate reams of flawless code. Gordon trusted nobody. He would use a different name on every magazine subscription—A. Gordon Letwin, B. Gordon Letwin, and so on—just to track down the source of any junk mail. He married a woman named Rose, and they adopted a baby pig that they treated like a member of the family. The pig grew to be seventy pounds or more, and would blast through its pig door into their home like a fullback going off tackle. Years later I heard that Gordon was taking it around with him on his Learjet.

Highlight (Yellow) | Location 1639

In June 1978, Intel introduced the 8086, one of the first 16-bit microprocessors and the next evolutionary step in personal computer technology. I was in regular contact with Intel and had the 8086 data sheet and instruction set well ahead of the official release. By then it was old hat for me to create development tools for chips from their specs, sight unseen.

Highlight (Yellow) | Location 1645

Though there was no 16-bit hardware on the horizon, I was determined not to wait as we had for the Altair. I set to work to simulate the 8086 on the PDP-10 and rewrote my macros for the larger instruction set. When the next-generation boxes materialized, we'd be ready.

Highlight (Yellow) | Location 1647

With Bill always worried about meeting expenses, we'd make commitments with little regard for our capacity to fulfill them. People had no choice but to work longer and harder—even Miriam Lubow, Bill's secretary, who sneaked in on weekends to do her filing without telling her husband. We maxed out what we could do on the school district's PDP-10 and switched to a faster one in Denver.

Highlight (Yellow) | Location 1668

When someone ran late on a job, he had a stock response: "I could code that in a weekend!"

Highlight (Yellow) | Location 1669

And if you hadn't thought through your position or Bill was just in a lousy mood, he'd resort to his classic put-down: "That's the stupidest fucking thing I've ever heard!"

Highlight (Yellow) | Location 1670

Good programmers take positions and stick to them, and it was common to see them square off in some heated disagreement over coding architecture. But it was tough not to back off against Bill, with his intellect and foot-tapping and body-rocking; he came on like a force of nature.

Highlight (Yellow) | Location 1672

irony was that Bill liked it when someone pushed back and drilled down with him to get to the best solution. He wouldn't pull rank to end an argument. He wanted you to overcome his skepticism, and he respected those who did. Even relatively passive people like Bob Wallace learned to stand their ground and match their boss decibel for decibel. They'd get right into his face: "What are you saying, Bill? I've got to write a compiler for a language we've never done before, and it needs a whole new set of runtime routines, and you think I can do it over the weekend? Are you kidding me?" I saw this happen again and again. If you made a strong case and were fierce about it, and you had the data behind you, Bill would react like a bluffer with a pair of threes. He'd look

down and mutter, “OK, I see what you mean,” then try to make up. Bill never wanted to lose talented people. “If this guy leaves,” he’d say to me, “we’ll lose all our momentum.”

Highlight (Yellow) | Location 1702

and the rainy days were a plus; they’d keep programmers from getting distracted. We agreed to finish out our lease and move home at the end of the year. Which is the story of how Seattle inherited what is now its second-largest employer. On Pearl Harbor Day, 1978, the Microsoft staff convened on the second floor of a shopping center for a group portrait. Despite a rare and raging snowstorm in Albuquerque that day, eleven of thirteen made it to Royal Frontier Studios. Ric Weiland was house-hunting in Seattle, and Miriam Lubow’s husband told her she’d be crazy to drive the three miles into town. (Miriam was the only employee who wouldn’t make the move with us, though she’d follow later on.)

Highlight (Yellow) | Location 1711

I had to leave two weeks before the others to set up the mainframe we’d bought for our software development. I scanned the map and saw that the shortest route went through Utah and Idaho, and then into Washington. I didn’t bother to check the forecast, and it was snowing like the dickens by the time I reached the Four Corners area in Utah. I was sliding all over the place in my rear-wheel-drive Monza. At one point I was listening to Earth, Wind & Fire when I spun clear off the road, which scared the heck out of me. I tend to be obstinate in adversity. I put chains on my tires. By the time I reached a mountainous stretch between Utah and Idaho called Dead Man’s Pass, the highway was one solid sheet of ice. I passed lines of semis that had either skidded off the pavement or swerved onto an escape ramp. Most sensible people would have turned back. But I white-knuckled it down that mountain, half-sure that I was going to shoot through the guardrail. When I finally reached Seattle (and it took close to a week), I sent word back that people should take the California route instead. They all had smooth sailing, except for Bill. He reportedly collected three speeding tickets, two of them from the same cop.

Highlight (Yellow) | Location 1905

As Bill noted in a 1993 interview for the Smithsonian: [The] question was, “Should we spread those products over to other 8-bit chips, like the 6502 that runs in [the Apple II]? Or, should we immediately move up and do 16-bit software?” And I said, “No, we are going to do 16-bit software.” Everybody was a little bit disappointed because it meant that we wouldn’t be able to sell onto these machines. That is when Paul invented the idea of the SoftCard, so that we could actually take our Intel software and run it on this machine, and, at the same time, go ahead and devote our resources to being way ahead of everybody else in developing software for the 8086. I had already been instituting the move to 16-bit software, but Bill wasn’t wrong about the SoftCard’s importance. Under the circumstances, I felt that our 64–36 partnership split was out of whack. Bill had set a precedent by claiming extra equity for his work on Altair BASIC, another exceptional contribution. Now it was time, I thought, to augment my share. A modest adjustment in the ratio seemed only right. But when I made my case, Bill would have none of it. “I don’t ever want to talk about this again,” he said. “Do not bring it up.” In that moment, something died for me. I’d thought that our partnership was based on fairness, but now I saw that Bill’s self-interest overrode all other considerations. My partner was out to grab as much of the pie as possible and hold on to it, and that was something I could not accept. I didn’t have it out with Bill at the time. I sucked it up and thought, OK...but one day I’m out of here.

Highlight (Yellow) | Location 2050

“We’ve got to focus,” he said. Then I told him about IBM’s move into the personal computer market. That was a different story, Charles agreed. We were so tight-lipped that we wouldn’t even mention the famous corporate acronym by name. We referred to our customer as HAL, after the computer in 2001: A Space Odyssey. (As movie buffs know, HAL is IBM moved one letter down the alphabet.) New levels of security were needed. When the PC prototype arrived at our office around Thanksgiving, it was closeted in a small, windowless backroom, under strict lock and key. Access was limited to a handful of people.

Highlight (Yellow) | Location 2120

Steve complemented Bill as a sounding board on business strategy, as I did on technical strategy. Bill remained the big-picture tactician, but Steve made us more disciplined and systematic. The two of them could get adversarial at times, with Steve’s arm-waving histrionics feeding into Bill’s pitiless dissections of what he thought everyone else was doing wrong. They were both ultracompetitive, super-high-IQ, maniacally relentless people with a tendency toward melodrama.

Highlight (Yellow) | Location 2139

Above all, we were after the brightest lights. A great programmer can outproduce an average one by ten to one; with a genius, the ratio might be fifty to one.

Highlight (Yellow) | Location 2152

Bill got so many speeding tickets that he had to hire the best traffic attorney in the state to defend him. He

Highlight (Yellow) | Location 2157

As a technology company grows, it must balance the need for innovation with the imperative to bolster existing products and keep the profits flowing. As Microsoft expanded far, far beyond the thirty-five programmers that once seemed like a pipe dream, it would get more and more difficult to keep that balance.

Highlight (Yellow) | Location 2180

IBM had no wish to support two sets of application software or to revisit the chaos of the 8-bit market. It priced our DOS at \$40 and CP/M-86 at \$240, a premium that reflected Kildall’s insistence on per-copy royalties. Once it became clear to users that IBM’s own software would run only on DOS, the independent software vendors ignored CP/M-86 and kept flocking to our operating system en masse. Our relationship with IBM made MS-DOS’s supremacy a fait accompli. The IBM PC was imperfect. It was expensive and looked corporate and soulless. But it was also the best personal computer of its time, with the most powerful processor, a superior keyboard, and reliable floppy disk drives. Executives didn’t wait for their IT officers to take the plunge into microcomputing; they went out, bought their PCs, and plugged them in. IBM had projected the sale of 250,000 units within five years, and they were off by a factor of ten. The PC marked the official close of the hobbyist era in computing,

Highlight (Yellow) | Location 2191

By 1984, the industry had converged around a single standard: MS-DOS twinned to the IBM specification. Legions of applications were developed to run on top of it. Hobbled by inferior hardware and limited software, other platforms could not compete. The TRS-80 was extinct by 1986; the Commodore 64 and Atari ST died a few years later. Only Apple survived, carving out a premium niche in the U.S. market that would hover around 10 percent.

Highlight (Yellow) | Location 2194

Down the road, Microsoft would suffer from the distraction of OS/2 until the company jettisoned the jointly created operating system in 1990 and signaled the end of its development partnership with IBM. Nevertheless, our original deal with Big Blue was critical both for Microsoft and for the industry as a whole. The PC phenomenon gave personal computers credibility in the business world. It became the launchpad for Windows and GUI

Highlight (Yellow) | Location 2197

Charles Simonyi, who has flown twice to the International Space Station, likes to use a rocket analogy. IBM gave Microsoft an essential boost before becoming a drag that we needed to shed.

Highlight (Yellow) | Location 2199

For me, the poignant postscript to this story was the fall of Digital Equipment Corporation, the company we idolized in our youth. Early on, its top management had resisted the shift to personal computers. As late as 1977, its president told the World Future Association, “There is no reason anyone would want a computer in their home.” Five years later, DEC swallowed its pride and released a well-designed microcomputer called the Rainbow 100, but shot itself in the foot by using the old 8-bit CP/M system. When DEC finally got around to implementing MS-DOS, the company required a customized version that was incompatible with thousands of PC applications, and it flopped. In the mainframe and minicomputer worlds, DEC had achieved small miracles by marching to its own drummer. If, ten years earlier, you had told us that IBM would forge a new path for microcomputers and that the maker of the PDP-10 would straggle in late and be unable to adapt, we would never have believed you.*

Highlight (Yellow) | Location 2278

I was reminded of that incident by a 1999 television movie called *Pirates of Silicon Valley*. In reimagining Microsoft in the late seventies and early eighties, the film depicts Bill as an übernerd and Steve Ballmer as a crude cheerleader, while I’m the bearded sidekick who loves gadgets and cracks wise now and then. Jobs, played by Noah Wyle, comes off as a charismatic but ruthless and mean-spirited jerk. The next time I ran into him, I asked him what he thought. And Jobs said, “I thought the guy who played me did a fantastic job.” He just didn’t care about what people thought of his public persona. Bill was different; he wanted to be viewed as tough but fair. He could be callous and rude, but he had a warmer, human side, too. And no one doubted that his excesses, for good or ill, were spontaneous. When Bill blew his stack at a meeting, it was never merely for

effect. As the late Bob Wallace once said, “The only stuff worth discussing was the stuff you didn’t agree about, so you tended mostly to argue with Bill.” Microsoft veterans knew the drill, but new employees could be traumatized. Bill would miss the cues from those who reacted with stony silence or cold frustration. He didn’t always notice when he pushed too hard.

Highlight (Yellow) | Location 2365

When I asked him whether two buttons might be better, he passionately lectured me: “You know, Paul, this is all about simplicity versus complexity. And nobody needs more than one button on a mouse.” I said, “But Steve, people have more than one finger, and there’s going to be things they might want to do with a right click, too.” Jobs dismissed my point with a shake of his head. He believed in making the entry-level experience as unintimidating as possible—and that there was usually one and only one correct way to do things. At Microsoft, we tried to balance simplicity with power. I considered the trade-off worthwhile if an extra feature made a program or device more functional.

Highlight (Yellow) | Location 2377

Today that extra button helps millions of Windows users gain access to context menus and a host of other convenient features. Postscript: In 2005, after twenty-two years of one-button worship, Apple relented and released the multibutton Mighty Mouse.

Highlight (Yellow) | Location 2543

As Bill told Tom Brokaw on an NBC special in 1995, technology was “a very scary business. If you fall behind technically, no matter how much your past success has been, it’s no guarantee that you’ll keep doing well in the future.” Within months of that appearance, in an internal memo that spread swiftly over the Internet, Bill pointed to Netscape Navigator as the main obstacle to Microsoft’s ascendance on the Internet “tidal wave,” which he now considered the key to the company’s future. And because Navigator didn’t require a richly endowed operating system, it was also a threat to Windows and Office, Microsoft’s core businesses. Leaders tend to stick with the style that made them successful. In Bill’s case, the tried-and-true formula was hyperaggression and supercompetitiveness. When your company becomes the industry leader, though, the game changes, and you need to ease off to avoid too much resentment from the rest of the ecosystem. But Bill couldn’t back down; that wasn’t in his DNA. He sent the same public message, over and over again: We take on all comers, and we clobber them. Just days before the Brokaw special, when the New York Times Magazine portrayed Microsoft on its cover as an eight-hundred-pound gorilla, it signaled something ominous. Bill’s hard-nosed approach, on top of Microsoft’s run of success, had provoked a backlash from competitors and their allies in the federal government. The antitrust campaign against Microsoft came to a head in 1997, when the Justice Department demanded a stiff fine against the company for alleged violations. A parallel case was advancing in the European Union.

Highlight (Yellow) | Location 2565

The Justice Department’s hired legal gun, David Boies, had pushed all of Bill’s buttons, and Bill took it from there. Rather than simply respond to Boies’s questions, he belittled them: “What do you mean by Internet

software?” Not only is that question ridiculous, but I’m going to explain just how ridiculous it is—and how clueless you are about the software business. Bill was sarcastic, combative, defensive, and contemptuous. I knew those traits well, but they were less than helpful on the stand. He might as well have told the Department of Justice (and by extension the judge) that the antitrust case was the stupidest thing he had ever heard. Anti-Microsoft sentiment became widespread and intense, and it cut Bill to the core. He’d been the darling of the business press, the crafty entrepreneur and technology genius. Now the media portrayed him as a bully who’d bent the rules and probably broken them.

Highlight (Yellow) | Location 2623

How did a company once at the forefront of technology and change fall so far behind? It’s a thorny question, with roots that go back decades, but I believe it boils down to three broad factors: scale, culture, and leadership.

Highlight (Yellow) | Location 2625

Microsoft got huge and failed to deal with the consequences. When I left the company, it had fewer than five hundred employees. By 1990, there were more than five thousand; by 2000, nearly forty thousand; today, more than ninety thousand. At that scale, cultural changes creep in unless you guard zealously against them. To avoid mediocrity, you need to be rigorous about weeding out underperformers. Microsoft hasn’t proven to be good at that. One executive recently told me, “I wish I could shoot every fourth one.”

Highlight (Yellow) | Location 2631

During 2009 and 2010, I had lunch with more than a dozen people who had recently left the company. They all said the same thing: too many semicompetent managers, too much in-house politics among the fiefdoms and silos of principal product lines. Windows Vista was the dead canary in the coal mine.

Highlight (Yellow) | Location 2638

Microsoft bounced back by finding a drill sergeant par excellence, Steven Sinofsky, to manage the development of Vista’s successor, Windows 7.

Highlight (Yellow) | Location 2639

When we began, our mission was narrowly defined as the microprocessor language company. Bill and I were programmers who developed software for other programmers and sold licenses to computer hardware manufacturers.

Highlight (Yellow) | Location 2641

Our DOS deal with IBM marked a departure from that safe home base. Once you start shipping operating systems, and then GUIs and word processors, your products go directly to end users.

Highlight (Yellow) | Location 2643

But it could be argued that it was never the company's forte to design products that made the consumer's heart beat faster. That history is currently reflected at Microsoft in the tension between selling to the end user and selling to what's known as "the enterprise," the server-based corporate market.

Highlight (Yellow) | Location 2648

when you're "the No. 1 wholesale seller of plumbing supplies," as the New York Times recently put it, innovation stops being organic. You may want to innovate, but it can be like trying to fight gravity.

Highlight (Yellow) | Location 2651

In the early years, we overlooked databases. Had the IBM opportunity not fallen into our lap, Microsoft might have been a footnote in operating systems; had Charles Simonyi not shown up at our door, we could easily have missed out on word processors, because no one in-house knew how to write one. In these markets and a number of others, Microsoft thrived as a fast follower, the company's MO from MS-DOS through Windows to Word and Excel.

Highlight (Yellow) | Location 2657

For a company with a leading market share and a bottomless war chest, it was the ideal approach: minimal risk for maximum return. Then the world changed (again). As content migrated to the Web, the pace of innovation accelerated. Fast following became more difficult than in the era of disk-based software. Today, for the most part, the best opportunities now lie where your competitors have yet to establish themselves, not where they're already entrenched.

Highlight (Yellow) | Location 2674

The new PC platform sparked killer apps like WordPerfect and Lotus 1-2-3, which owned their respective markets up until they failed to adapt to the next big advance in access, the graphic user interface. When Microsoft's Windows and superior GUI-powered applications evolved, it was game over.

Highlight (Yellow) | Location 2681

mobile platforms, which are rapidly becoming the principal delivery point for low- to-medium-intensity computing and Web content consumption.

Highlight (Yellow) | Location 2688

Today's robust iPhones and Droids are the products of high-speed communications, low-cost manufacturing, and superfast microprocessors. Apple and Google have beaten Microsoft to the mobile punch because they've been more alert in developing new and innovative platforms. They've done a better job of following the chips. Increasing Computational Performance and Decreasing Hardware Costs

Highlight (Yellow) | Location 2693

Apple is the ultimate auteur company with the most fervent cult following in the business. The world of Jobs offers limited options (there's only one basic flavor of iPhone),

Highlight (Yellow) | Location 2697

cloud-oriented Google, which takes a different approach: start with limited functionality, copy the leader, bring in apps from everywhere, iterate like mad. In essence, it has mastered Microsoft's old strategy of fast following for the mobile, Web-based era.

Highlight (Yellow) and Note | Location 2710

It appears that tablets are poised to render physical books, magazines, and newspapers obsolete within the next twenty years. As an inveterate book lover, I find the prospect sad but inevitable.

I'm rereading his book on a kindle...

Highlight (Yellow) | Location 2715

I often hear people saying, "I don't like it personally, but my kids are perfectly happy typing on their iPhone."

Highlight (Yellow) | Location 2719

Here's what the death knell for the personal computer will sound like: Mainly I use my phone/pad, but I still use my PC to write long e-mails and documents. Most people aren't there yet, but that's where we're headed. If Microsoft fails to catch up in mobile, in other words, it's in for a long, slow slide.

Highlight (Yellow) | Location 2735

squeezed Altair BASIC into four thousand bytes, no

Highlight (Yellow) | Location 2749

By 1990, at age thirty-seven, I'd become a billionaire. By 1996, I'd be one ten times over.

Highlight (Yellow) and Note | Location 2779

As we wrapped up, Weinberg said he would take \$65 million. Ten minutes later, we had a handshake deal. I was thirty-five years old and the youngest owner in major-league sports.

Fall '87

Highlight (Yellow) | Location 2827

I've tried to strike a balance as team owner, to be involved and accountable while preserving my executives' freedom to shape the roster. My job is oversight, not execution. While I sign off on trades or free agents, I've rarely overruled my basketball people's decisions. But I'm not shy about steering the discussion or pushing deeper if something doesn't make sense to me.

Highlight (Yellow) | Location 2859

I've seen just one other person up close who compared to him, who wanted not only to beat you but to crush you if he could. Those two stood apart for raw competitiveness: Michael Jordan and Bill Gates.
